

RevoFin

Yearly Analytical Report



Disclaimer

Please note that RevoFin is a fictional company used for the purposes of this presentation. Any references to its services or data analysis methods are purely illustrative. This scenario is designed to showcase the skills of a data analyst and is not linked to any real financial institution. Any resemblance to actual companies is coincidental.

Background and Business Problems

RevoFin requires continuous monitoring of its loan portfolio to ensure:

- Strong financial health
- Controlled credit risk
- Sustainable lending growth

However, portfolio risk does not only depend on total exposure, but also on:

- Borrower quality
- Loan characteristics
- Time-based (cohort) performance

Without proper analytical evaluation, risk deterioration may go undetected.



Objectives & Analytical Scope

This analysis aims to:

- Present the **outstanding loan amount (OS)** of the portfolio
- Highlight key **risk metrics (ENR & TKB30)**
- Perform **cohort-based risk evaluation**
- Conduct **deep-dive profiling** using:
 - Loan characteristics (purpose, interest rate)
 - Customer attributes (income, employment, home ownership)
- Demonstrate **SQL-based analytical capability**



Data & Methodology

Data Sources

- **Loan Table:** funded amount, issue year, loan status, interest rate, purpose
- **Customer Table:** income, employment length, home ownership, state

Key Data Preparation

- Cleaned `customer_id` using regex extraction
- Removed non-portfolio statuses (e.g., Charged Off)
- Defined portfolio as:
 - Current
 - In Grace Period
 - Late (16–30 days)
 - Late (31–120 days)
 - Default

Key Metrics Defined

- **OS (Outstanding Amount):** Total funded amount of active loans
- **ENR (Earning Outstanding):** OS excluding Default loans
- **DPD > 30:** Late (31–120) + Default
- **TKB30:**

$$\text{TKB30} = 1 - (\text{OS of loan with DPD} > 30) / (\text{Portfolio OS})$$

Techniques Used

- Aggregation, Cohort Analysis, Window Functions
- Benchmark comparison vs Portfolio Average
- Advanced SQL CTE pipelines

Portfolio Size & Risk Overview

Portfolio Scale

- **Total Customers:** 176,087
- **Total OS:** 28.01 Billion
- **Total ENR:** 28.01 Billion

ENR $\approx$ OS $\rightarrow$ Almost all exposure is still generating income

Portfolio Risk (TKB30)

- **Portfolio TKB30:** 98.11%
- Only **1.89%** of OS is already >30 days past due

Indicates a **generally healthy but actively monitored portfolio**

Cohort Analysis

Cohort Performance by Issue Year

- Loans grouped by **issue year**
- OS, ENR, and TKB30 calculated per cohort

Key Findings

- Lowest TKB30 Cohort: 2012**
(≈96.25%)
- Highest TKB30 Cohort: 2019**
(≈98.83%)
- Clear improvement in repayment quality over time

Underwriting quality has **structurally improved across newer cohorts**

Year	OS	ENR	TKB30
2012	2,346,175	2,346,175	0.96
2013	23,986,075	23,951,075	0.97
2014	128,947,375	128,933,450	0.97
2015	84,824,900	84,824,900	0.97
2016	549,214,275	549,068,925	0.98
2017	486,738,000	486,738,000	0.97
2018	738,165,450	738,164,450	0.99
2019	786,856,600	786,856,600	0.99 ⁷

Deep Dive Insights: Financial Stability

Home Ownership Rate

- Low TKB30: **57.4%**
- Portfolio Avg: **60.7%**
- High TKB30: **61.2%**

Employment Length

- All cohorts $\approx$ **6 years**
- No significant differentiation

Average Annual Income

- Low TKB30: **72,334**
- Portfolio Avg: **78,382**
- High TKB30: **85,144**

- ✓ Lower income and lower asset ownership strongly drive risk
- ✗ Employment length is not a key risk discriminator

Deep Dive Insights: Concentration & Pricing

Geographic Concentration (Top State: CA)

- Low TKB30: **15.2%**
- Portfolio Avg: **13.1%**
- High TKB30: **13.4%**

Loan Purpose Concentration

- Dominant purpose: **Debt Consolidation**
Low TKB30: **64.3%**
- Portfolio Avg: **55.4%**
- High TKB30: **54.5%**

Interest Rate

- Low TKB30: **14.10%**
- Portfolio Avg: **13.11%**
- High TKB30: **12.89%**

Weakest cohort shows:

- **Higher geographic concentration**
- **Heavy exposure to debt consolidation**
- **Higher risk-based pricing**

Key Insights Summary

What Drives the Weakest Cohort (2012)?

- Lower income
- Lower home ownership
- Higher geographic concentration
- Heavy exposure to debt consolidation loans
- Higher interest rates
- **Employment length not a risk driver**

What Characterizes the Strongest Cohort (2019)?

- Higher income
- Higher home ownership
- More diversified loan purpose
- Lower interest rates

Structural borrower quality is the **primary risk differentiator**, not loan tenure.

Recommendations

Underwriting & Risk Management Recommendations

1. **Tighten income screening** for debt consolidation loans
2. Apply **geographic concentration limits** in high-risk states
3. Strengthen **asset-based (home ownership) screening**
4. Enhance **early-warning monitoring** for high-interest borrowers
5. Maintain **risk-based pricing**, but combine with stricter approval thresholds

Expected Impact

- Lower future delinquency
- More diversified portfolio
- Higher future cohort TKB30
- Stronger long-term portfolio resilience

Executive Conclusion

- RevoFin's portfolio is **financially strong** with TKB30 = **98.11%**
- Risk quality has **improved across cohorts**
- The weakest cohort shows **clear structural financial weaknesses**
- Data-driven underwriting refinement can **further stabilize portfolio risk**





Thank you

Number of Customers, OS, ENR

```
WITH loan_cleaned AS (  
  SELECT  
    ARRAY_TO_STRING(REGEXP_EXTRACT_ALL(customer_id, r"[a-zA-Z0-9]"), "") AS clean_customer_id,  
    loan_status,  
    funded_amount  
  FROM `next-porto.fip.loan`  
  WHERE loan_status IN ('Current', 'In Grace Period', 'Late (16-30 days)', 'Late (31-120 days)', 'Default')  
)  
  
SELECT  
  COUNT(DISTINCT clean_customer_id) AS number_of_customers,  
  SUM(funded_amount) AS outstanding_amount_os,  
  SUM(  
    CASE  
      WHEN loan_status <> 'Default' THEN funded_amount  
      ELSE 0  
    END  
  ) AS current_enr  
FROM loan_cleaned;
```

TKB30

```
WITH loan_cleaned AS (  
  SELECT  
    ARRAY_TO_STRING(REGEXP_EXTRACT_ALL(customer_id, r"[a-  
zA-Z0-9]"), "") AS clean_customer_id,  
    loan_status,  
    funded_amount  
  FROM `next-porto.fip.loan`  
  WHERE loan_status IN ('Current', 'In Grace Period',  
    'Late (16-30 days)', 'Late (31-120 days)', 'Default')  
)
```

```
agg AS (  
  SELECT  
    SUM(funded_amount) AS portfolio_os,  
    SUM(  
      CASE  
        WHEN loan_status IN ('Late (31-120 days)',  
    'Default')  
        THEN funded_amount  
        ELSE 0  
      END  
    ) AS os_dpd_gt_30  
  FROM loan_cleaned  
)
```

```
SELECT  
  1 - (os_dpd_gt_30 / portfolio_os) AS tkb30  
FROM agg;
```


YEARLY COHORT

```
WITH loan_cleaned AS (
  SELECT
    ARRAY_TO_STRING(REGEXP_EXTRACT_ALL(customer_id, r"[a-zA-Z0-9]"), "") AS clean_customer_id,
    loan_status,
    funded_amount,
    CAST(issue_year AS INT64) AS cohort_year
  FROM `next-porto.fip.loan`
  WHERE loan_status IN ('Current', 'In Grace Period',
    'Late (16-30 days)', 'Late (31-120 days)', 'Default')
),
```

```
cohort_agg AS (
  SELECT
    cohort_year,
    SUM(funded_amount) AS os,
    SUM(CASE WHEN loan_status <> 'Default' THEN
      funded_amount ELSE 0 END) AS enr,
    SUM(CASE WHEN loan_status IN ('Late (31-120
      days)', 'Default') THEN funded_amount ELSE 0 END) AS
    os_dpd_gt_30
  FROM loan_cleaned
  GROUP BY cohort_year
)
```

```
SELECT
  cohort_year,
  os,
  enr,
  1 - (os_dpd_gt_30 / os) AS tkb30
FROM cohort_agg
ORDER BY cohort_year;
```

COHORT ANALYSIS (HOME OWNERSHIP)

```
WITH loan_base AS (
  SELECT
    ARRAY_TO_STRING(REGEXP_EXTRACT_ALL(l.customer_id, r"[a-zA-Z0-9]"), "") AS clean_customer_id,
    CAST(l.issue_year AS INT64) AS cohort_year,
    l.int_rate,
    l.purpose
  FROM `next-porto.fip.loan` l
),
```

```
loan_tagged AS (
  SELECT
    clean_customer_id,
    int_rate,
    purpose,
    CASE
      WHEN cohort_year = 2012 THEN 'Low_TKB30_Cohort'
      WHEN cohort_year = 2019 THEN 'High_TKB30_Cohort'
      ELSE 'Other_Cohorts_Avg'
    END AS cohort_group
  FROM loan_base
),
```

```
joined AS (
  SELECT
    lt.cohort_group,
    lt.int_rate,
    lt.purpose,
    c.annual_inc,
    c.emp_length,
    c.home_ownership,
    c.addr_state
  FROM loan_tagged lt
  JOIN `next-porto.fip.customer` c
  ON lt.clean_customer_id =
    ARRAY_TO_STRING(REGEXP_EXTRACT_ALL(c.customer_id, r"[a-zA-Z0-9]"), "")
)
```

```
SELECT
  cohort_group,
  AVG(CASE WHEN home_ownership IN ('OWN', 'MORTGAGE') THEN 1 ELSE
0 END) AS home_ownership_rate
FROM joined
GROUP BY cohort_group;
```

COHORT ANALYSIS (AVG ANNUAL INCOME)

```
WITH loan_base AS (
  SELECT
    ARRAY_TO_STRING(REGEXP_EXTRACT_ALL(l.customer_id, r"[a-zA-Z0-9]"), "") AS clean_customer_id,
    CAST(l.issue_year AS INT64) AS cohort_year,
    l.int_rate,
    l.purpose
  FROM `next-porto.fip.loan` l
),
```

```
loan_tagged AS (
  SELECT
    clean_customer_id,
    int_rate,
    purpose,
    CASE
      WHEN cohort_year = 2012 THEN 'Low_TKB30_Cohort'
      WHEN cohort_year = 2019 THEN 'High_TKB30_Cohort'
      ELSE 'Other_Cohorts_Avg'
    END AS cohort_group
  FROM loan_base
),
```

```
joined AS (
  SELECT
    lt.cohort_group,
    lt.int_rate,
    lt.purpose,
    c.annual_inc,
    c.emp_length,
    c.home_ownership,
    c.addr_state
  FROM loan_tagged lt
  JOIN `next-porto.fip.customer` c
  ON lt.clean_customer_id =
    ARRAY_TO_STRING(REGEXP_EXTRACT_ALL(c.customer_id, r"[a-zA-Z0-9]"), "")
)
```

```
SELECT
  cohort_group,
  AVG(annual_inc) AS avg_income,
FROM joined
GROUP BY cohort_group;
```

COHORT ANALYSIS (EMPLOYMENT LENGTH)

```
WITH loan_base AS (
  SELECT
    ARRAY_TO_STRING(REGEXP_EXTRACT_ALL(l.customer_id, r"[a-zA-Z0-9]"), "") AS clean_customer_id,
    CAST(l.issue_year AS INT64) AS cohort_year,
    l.int_rate,
    l.purpose
  FROM `next-porto.fip.loan` l
),
```

```
loan_tagged AS (
  SELECT
    clean_customer_id,
    int_rate,
    purpose,
    CASE
      WHEN cohort_year = 2012 THEN 'Low_TKB30_Cohort'
      WHEN cohort_year = 2019 THEN 'High_TKB30_Cohort'
      ELSE 'Other_Cohorts_Avg'
    END AS cohort_group
  FROM loan_base
),
```

```
joined AS (
  SELECT
    lt.cohort_group,
    lt.int_rate,
    lt.purpose,
    c.annual_inc,
    c.emp_length,
    c.home_ownership,
    c.addr_state
  FROM loan_tagged lt
  JOIN `next-porto.fip.customer` c
  ON lt.clean_customer_id =
    ARRAY_TO_STRING(REGEXP_EXTRACT_ALL(c.customer_id, r"[a-zA-Z0-9]"),
    "")
)

SELECT
  cohort_group,
  AVG(
    SAFE_CAST(REGEXP_EXTRACT(emp_length, r"\d+") AS INT64)
  ) AS avg_emp_years
FROM joined
GROUP BY cohort_group;
```

COHORT ANALYSIS (STATE CONCENTRATION) (1)

```

WITH loan_base AS (
  SELECT
    ARRAY_TO_STRING(
      REGEXP_EXTRACT_ALL(l.customer_id, r"[a-zA-Z0-9]"),
      ""
    ) AS clean_customer_id,
    CAST(l.issue_year AS INT64) AS cohort_year
  FROM `next-porto.fip.loan` AS l
  WHERE l.loan_status IN (
    'Current',
    'In Grace Period',
    'Late (16-30 days)',
    'Late (31-120 days)',
    'Default'
  )
),

```

```

loan_tagged AS (
  SELECT
    clean_customer_id,
    CASE
      WHEN cohort_year = 2012 THEN 'Low_TKB30_Cohort'
      WHEN cohort_year = 2019 THEN 'High_TKB30_Cohort'
      ELSE 'Other_Cohorts_Avg'
    END AS cohort_group
  FROM loan_base
),

joined AS (
  SELECT
    lt.cohort_group,
    c.addr_state AS addr_state
  FROM loan_tagged AS lt
  JOIN `next-porto.fip.customer` AS c
    ON lt.clean_customer_id =
      ARRAY_TO_STRING(
        REGEXP_EXTRACT_ALL(c.customer_id, r"[a-zA-Z0-9]"),
        ""
      )
),

```

COHORT ANALYSIS (STATE CONCENTRATION) (2)

```
state_dist AS (  
  SELECT  
    cohort_group,  
    addr_state,  
    COUNT(*) AS n  
  FROM joined  
  GROUP BY cohort_group, addr_state  
) ,  
  
state_share_cte AS (  
  SELECT  
    cohort_group,  
    addr_state,  
    n,  
    n / SUM(n) OVER (PARTITION BY cohort_group) AS state_share  
  FROM state_dist  
) ,
```

```
ranked AS (  
  SELECT  
    cohort_group,  
    addr_state,  
    n,  
    state_share,  
    ROW_NUMBER() OVER (  
      PARTITION BY cohort_group  
      ORDER BY state_share DESC  
    ) AS rn  
  FROM state_share_cte  
)  
  
SELECT  
  cohort_group,  
  addr_state AS top_state,  
  n AS loans_in_top_state,  
  state_share AS top_state_share  
FROM ranked  
WHERE rn = 1  
ORDER BY cohort_group;
```

COHORT ANALYSIS (LOAN PURPOSE) (1)

```
WITH loan_base AS (
  SELECT
    ARRAY_TO_STRING(
      REGEXP_EXTRACT_ALL(l.customer_id, r"[a-zA-Z0-9]"),
      ""
    ) AS clean_customer_id,
    CAST(l.issue_year AS INT64) AS cohort_year,
    l.purpose
  FROM `next-porto.fip.loan` AS l
  WHERE l.loan_status IN (
    'Current',
    'In Grace Period',
    'Late (16-30 days)',
    'Late (31-120 days)',
    'Default'
  )
),
```

```
loan_tagged AS (
  SELECT
    clean_customer_id,
    purpose,
    CASE
      WHEN cohort_year = 2012 THEN 'Low_TKB30_Cohort'
      WHEN cohort_year = 2019 THEN 'High_TKB30_Cohort'
      ELSE 'Other_Cohorts_Avg'
    END AS cohort_group
  FROM loan_base
),
```

```
purpose_dist AS (
  SELECT
    cohort_group,
    purpose,
    COUNT(*) AS n
  FROM loan_tagged
  GROUP BY cohort_group, purpose
),
```


COHORT ANALYSIS (LOAN PURPOSE) (2)

```
purpose_share_cte AS (  
  SELECT  
    cohort_group,  
    purpose,  
    n,  
    n / SUM(n) OVER (PARTITION BY cohort_group) AS  
    purpose_share  
  FROM purpose_dist  
)
```

```
ranked AS (  
  SELECT  
    cohort_group,  
    purpose,  
    n,  
    purpose_share,  
    ROW_NUMBER() OVER (  
      PARTITION BY cohort_group  
      ORDER BY purpose_share DESC  
    ) AS rn  
  FROM purpose_share_cte  
)  
  
SELECT  
  cohort_group,  
  purpose AS top_loan_purpose,  
  n AS loans_in_top_purpose,  
  purpose_share AS top_purpose_share  
FROM ranked  
WHERE rn = 1  
ORDER BY cohort_group;
```

COHORT ANALYSIS (AVG INTEREST RATE)

```
WITH loan_base AS (
  SELECT
    ARRAY_TO_STRING(REGEXP_EXTRACT_ALL(l.customer_id, r"[a-zA-Z0-9]"), "") AS clean_customer_id,
    CAST(l.issue_year AS INT64) AS cohort_year,
    l.int_rate,
    l.purpose
  FROM `next-porto.fip.loan` l
),
```

```
loan_tagged AS (
  SELECT
    clean_customer_id,
    int_rate,
    purpose,
    CASE
      WHEN cohort_year = 2012 THEN 'Low_TKB30_Cohort'
      WHEN cohort_year = 2019 THEN 'High_TKB30_Cohort'
      ELSE 'Other_Cohorts_Avg'
    END AS cohort_group
  FROM loan_base
),
```

```
joined AS (
  SELECT
    lt.cohort_group,
    lt.int_rate,
    lt.purpose,
    c.annual_inc,
    c.emp_length,
    c.home_ownership,
    c.addr_state
  FROM loan_tagged lt
  JOIN `next-porto.fip.customer` c
  ON lt.clean_customer_id =
    ARRAY_TO_STRING(REGEXP_EXTRACT_ALL(c.customer_id, r"[a-zA-Z0-9]"), "")
)
```

```
SELECT
  cohort_group,
  AVG(int_rate) AS avg_interest_rate
FROM joined
GROUP BY cohort_group;
```